UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
COURSE CODE	:	**CSC429**
EXAMINATION	:	**JULY 2024**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (10 Questions)
   PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A

QUESTION 1

The Difference Engine was a mechanical computer designed by Charles Babbage in the 19th century. It was one of the earliest examples of a mechanical calculator, built to automatically compute mathematical tables for naval navigation.

a) What device was the successor of the Difference Engine?

(1 mark)

b) Describe the difference in function between the Difference Engine and Mechanical Calculating Machine.

(2 marks)

c) Briefly explain the available technology that existed in the computer generation of Difference Engine.

(2 marks)

QUESTION 2

John von Neumann was credited with introducing the Von Neumann architecture, a pioneering concept in computer design. Additionally, he contributed to the development of the two's complement numbering system.

a) What is the name of the machine that John Von Neumann himself invented?

(1 mark)

b) Why was the two's complement numbering system introduced?

(2 marks)

c) Name **TWO (2)** memory contents that are allowable in the Von Neumann architecture.

(2 marks)

QUESTION 3

a) Discuss **TWO (2)** reasons why the binary numbering system is used instead of a decimal numbering system in computer architecture.

(4 marks)

b) Estimate the number of vacuum tubes needed to represent the value of 79 using the binary representation. Show the workflow.

(1 mark)

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QUESTION 4

a) Solve $246_8 - 1A_{16}$ using the sign and magnitude method. The answer must be in byte form. Show the workflow.

(2 marks)

b) Calculate $-3E_{16} + 45_8$ using 1's and 2's complement method. The answer must be in byte form. Show the workflow.

(3 marks)

QUESTION 5

a) Convert 204.9375_{10} to a floating-point number in IEEE single precision format in hexadecimal value form. Show the workflow.

(5 marks)

b) Convert C248 $A000_H$ in IEEE single precision floating point number to decimal value form. Show the workflow.

(5 marks)

QUESTION 6

a) Discuss **TWO (2)** types of bus timing.

(4 marks)

b) Describe **THREE (3)** Central Processing Unit (CPU) components.

(6 marks)

QUESTION 7

Given that the size of the data bus is 32 bits, the size of the address bus is 8 bits, and the size of the control bus is 4 bits.

a) Determine the size of the word and provide relevant justification.

(2 marks)

b) Determine the size of the register and provide relevant justification.

(2 marks)

c) Discuss what will happen if system interconnection is blocked from the perspective of the machine cycle.

(4 marks)

d) Elaborate the role of CPU interconnection.

(2 marks)

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QUESTION 8

a) Elaborate **THREE (3)** types of replacement algorithm.

(6 marks)

b) List **TWO (2)** types of mapping that require a replacement algorithm.

(2 marks)

c) Discuss the reason why the replacement algorithm needs to be implemented.

(2 marks)

QUESTION 9

Figure 1 illustrates the icon of one of the interfaces to the external devices.

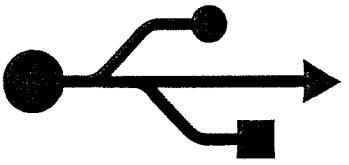


Figure 1: Interface to the external devices

a) Identify the type of external device interface based on the icon in Figure 1.

(2 marks)

b) The external devices interface in (a) can have a "hot connection". Define hot connection.

(2 marks)

c) Discuss the **THREE (3)** input-output techniques available in a computer system and elaborate on the Central Processing Unit's involvement in each technique.

(6 marks)

QUESTION 10

a) List the **FOUR (4)** classifications of parallel processors and provide examples for each.

(8 marks)

b) Discuss **ONE (1)** advantage of using cluster computer architecture.

(2 marks)

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PART B

QUESTION 1

Question 1 is based on Figure 2 and Table 1.

$$y = A(A + 2B / C - D^2)$$

Figure 2

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register ***A***, ***B, C,*** and ***D*** cannot be modified, and registers ***x, y***, and ***z*** can be used as temporary registers. However, the final answer must be stored in Register ***y***.

Table 1: Machine Instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	

(20 marks)

END OF QUESTION PAPER

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